Proceedings of IMECE '03:
2003 ASME International Mechanical Engineering Congress & Exposition
Washington, D.C., November 16-21, 2003

IMECE 2003-43865

HEAT TRANSFER IN WATER-COOLED SILICON CARBIDE MILLI-CHANNEL HEAT SINKS FOR HIGH POWER ELECTRONIC APPLICATIONS

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ABSTRACT

Heat transfer and fluid flow in a novel class of water-cooled milli-channel heat sinks are investigated. The heat sinks are manufactured using an extrusion freeform fabrication (EFF) rapid prototyping technology and a water-soluble polymer material. EFF permits the fabrication of geometrically complex, three-dimensional structures in non-traditional materials. Silicon carbide, SiC, is TEC-matched to silicon and is an ideal material for heat exchangers that will be mounted directly to heat dissipating electronic packages. This paper presents experimental results on the heat transfer and flow in small SiC heat exchangers with multiple rows of parallel channels oriented in the flow direction. Rectangular heat exchangers with 3.2 cm x 2.2 cm planform area and varying thickness, porosity, number of channels, and channel diameter were fabricated and tested. Overall heat transfer and pressure drop coefficients in single-phase flow regimes are presented and analyzed. The per-channel Reynolds number places the friction coefficients in the developing to developed hydrodynamic regime, and showed excellent agreement with laminar theory. The overall heat transfer coefficients for a single row SiC heat exchanger compared favorably with a validation heat exchanger fabricated from copper, however the heat transfer coefficient in multiple row heat sinks did not agree well with the laminar theory.

INTRODUCTION

With the trend towards increasing levels of integration in the IC industry, heat-sinking technologies with a higher level of performance are required to meet the elevated power dissipation requirements in electronic devices. With air as a working fluid, it is increasingly difficult to design cost-effective heat sinks that can reject over 100 W/cm^2 [1]. To this end, compact microchannel structures implementing higher conductivity fluids such as water have received renewed attention as a possible cooling alternative. Single row microchannels etched directly into the backs of silicon wafers were first shown to be effective by Tuckerman and Pease [2] in which a maximum of 790 W/cm^2 was rejected with a rise in water temperature of 71 C. While the cost of manufacturing microfabricated heat sinks currently prohibits application in production level electronics, the study showed that microchannel structures are well suited to the task of cooling electronic devices.

Following the pioneering work of Tuckerman and Pease, considerable research has been conducted on microchannel heat sinks. Much of this research has focused on single rows of thin, deep channels, commonly analyzed with the fin approach [3]. The fin approach has more recently found application in analysis of foam structures [4] and other low aspect ratio structures. The accuracy of the fin approach is limited by several simplifying assumptions, including fluid temperature uniformity in the direction perpendicular to the fluid flow, one-dimensional heat transfer in the fins, and constant heat transfer coefficient. Thus, the range of accuracy of the fin approach in modeling micro-channel hast sinks is restricted. [5].

More recently, the reduction of relevant dimensions in classical flow structures found in microchannel heat exchangers has shown that they are amenable to investigation using a porous media approach. In a single equation porous media model, the fluid and solid properties and geometry are no longer considered separately, but are considered to have combined, effective thermal properties. In a two-equation model, the volume averaged equations for the fluid and solid phases are connected by a heat transfer coefficient. Koh and Colony [6] and Kim et al [7,8] have approached the problem using either one or two equation models. This approach is well suited to channels with low aspect ratios such as foams or multiple rows of small straight channels.

To a large extent, research into micro- and milli-scale heat exchangers have been centered around heat sinks fabricated from a highly thermally conductive solid, such as copper or silicon, with rows of small channels fabricated into the surfaces by precision machining or chemical etching. High solid conductivity is particularly important in multiple row structures, as the amount of heat removed by any given row can be large. A highly conductive medium increases heat conduction into subsequent layers where it can be transferred to the fluid flow. This provides an advantage to multiple row heat sinks. In single-phase conditions, single row heat sinks exhibit large stream-wise increase in the water temperature due to high heat fluxes. This promotes a stream-wise increase in the temperature of the dissipating device [9]. This temperature increase is detrimental to temperature-sensitive devices such as microprocessors, which must operate below specified temperatures, and contributes to stresses arising from TEC mismatch. Vafai and Zhu [10] showed that a two-layer counterflowing micro-channel structure reduced the stream-wise temperature rise along the device surface, compared to that of the one-layered heat sink.

This paper presents results of an initial experimental study of the single phase heat transfer and pressure drop characteristics of compact, single and multi-row Silicon Carbide heat exchangers manufactured by an extrusion freeform fabrication method. This method allows the fabrication of heat sinks from non-traditional materials containing optimal number of rows, hole diameter, porosity, and other properties. The SiC implemented in this study has a thermal conductivity of approximately 15 W/m·C. While not an ideal conductor, SiC has a thermal expansion coefficient closely matched to that of silicon, allowing the exchanger to be more easily integrated directly into the packaging solution. The favorable TEC characteristic of these exchangers, when paired with the ease of manufacturing multiple row structures, has opened the door towards the development of integrated liquid cooled thermal solutions for high heat flux electronics.

NOMENCLATURE

a	= half width of channel, (m)
A	= area (m^2)
b	= half width of unit cell (m)
c_p	= specific heat of water (J/kg·C)
d	= channel diameter (m)
f	= friction factor
Gz	= Graetz number
H	= sample thickness (m)
k_e	= effective thermal conductivity of sample (W/m·C)
k_w	= water thermal conductivity (W/m·C)
L	= sample length (m)
$\dot{m}$	= volumetric flow rate flow rate (mL/min)
N	= number of channels
Nu	= Nusselt number
P	= pressure (Pa)
Pr	= Prandtl number
q	= heat flow (W)
R	= thermal resistance (C/W)
Re	= Reynolds number
T	= temperature (C)
u	= mean channel velocity (m/s)
U	= overall heat transfer coefficient (w/m^2·C)
w	= sample width (m)
x	= streamwise coordinate
x^+	= non-dimensional distance
y	= normal coordinate
Greek	
Δ	= delta, change in property
μ	= dynamic viscosity at mean temperature (N·s/m^2)
π	= pi
ρ	= fluid density (kg/m^3)
Φ	= porosity
Subscripts	
base	= based on contact area
d	= based on hydraulic diameter
e	= effective
h	= based on sample height
in	= inlet
m	= meter bar
out	= outlet
o-i	= outlet minus inlet
s	= solid or surface, depending on context
s-w	= log mean, surface to wall
y	= y direction, normal to flow
w	= water
wet	= based on wetted internal area

EXPERIMENTS

Test Samples and Instrumentation

Tests were performed on six SiC samples each measuring approximately 3.2 cm x 2.2 cm in planform area with varying thickness, H, number of channels, N, channel diameter, d, and number of channel rows. In addition, a copper validation sample with a single row of six two-millimeter diameter holes was fabricated for comparison. Figure 1 illustrates the nomenclature used in describing the test samples. The properties of the samples tested are shown in Table 1. Each of the samples was manifolded with 0.5 inch thick G-10, a low conductivity (0.288 W/m·C) glass/epoxy composite, allowing the sample to be flowed while minimizing heat lost through conduction. Pressure taps with a 0.08 cm ID were inserted 2 to 3 mm upstream and downstream of the exchanger inlet and exit respectively. Measurement of hydrodynamic pressure drop across a sample was performed by means of simple U-tube manometer. Figure 2 illustrates the sample with a manifold as configured for the heat transfer and pressure drop measurements. Type K, 36 gauge, thermocouples were inserted in the flow allowing direct measurement of the mixed mean temperature approximately two cm from the heat exchanger entry and exit.

Water flow was delivered to the heat sink via a Neslab RTE 7 constant temperature recirculator capable of absorbing 500 Watts at 20 C. Downstream of the recirculator outlet, flow metering was achieved with a bank of valved, correlated rotometers, with ranges from 1 to 1000 ml/. From the flow meters, water flowed into the inlet of the milli-channel sample. The fluid exiting the heat sink was collected at the outlet manifold and returned to the recirculator reservoir at ambient pressure to be chilled for recirculation.

Heat Delivery and Measurement

Heat was supplied symmetrically via a matched set of copper metering bars measuring 6.35 cm x 6.35 cm in cross section at the "head", which tapers to 2.54 cm x 2.54 cm in cross section at its contact with the sample. The bars contacted the heat exchanger external surfaces through a matching opening in the G-10 manifold. Three Incoloy cartridge heaters, each capable of developing 500 watts at 120 AC volts, were embedded in the "head" of each bar. Each set of heaters was driven via pulse width modulation of a standard 120 volt AC signal by a Fugi PXR-4 temperature controller and solid-state-relay combination. A series of five, 36 gauge, type K thermocouples were embedded in the meter bar at 0.25 inch intervals beginning at 0.125 inches from the contact surface. Heat flow was determined by measurement of the temperature gradient along the meter bar, and using the one-dimensional conduction law:

$$q = A_m k_m \Delta T / \Delta x \tag{1}$$

In all instances the temperature gradient was seen to be linear in the meter bar, validating the use of the one dimensional conduction equation. A high conductivity interface compound was applied to all contact surfaces to ensure evenly distributed thermal contact while allowing for the placement of three additional 36 gauge, type K thermocouples on the sample surface to estimate surface temperature. The heated end of the metering bars were attached to a precision platen die set and actuated by a pneumatic ram. The apparatus was designed for measuring interface resistance with one heated and one cooled metering bar. All thermocouples were terminated in an isothermal zone box and temperatures were measured with a Fluke Hydra 2620-A data acquisition unit.

A typical test was conducted as follows: After insertion of a sample between metering bars, a 200 psi load was applied to insure proper contact of surface thermocouples and to minimize contact resistance. The flow loop was engaged at maximum flow rate, commonly 500ml/min, and the heating was applied so as to establish a heater temperature of 90 C. The system was allowed to reach thermal equilibrium for a period of at least ten minutes or until surface temperatures changed by no more than 0.1 C in five minutes. Following collection of thermal and pressure data, the flow rate was systematically lowered and data acquisition was repeated.

DATA REDUCTION

Thermal Characterization.

The derived thermal metrics are stated in terms of either wet or base surface areas. A_{wet} is the cross sectional heated perimeter of the wetted channels:

$$A_{wet} = p\, d w_m N \tag{2}$$

A_{base} is the total outer heated area:

$$A_{base} = 2 w w_m \tag{3}$$

where w is the width of the sample and w_m is the width of the metering bar, not the entire sample length, L. The overall heat transfer coefficient from the SiC to the water is calculated as shown by Incropera and DeWitt[11]:

$$U = \frac{q}{A \Delta T_{s-w}} \tag{4}$$

where A can be either A_{wet} or A_{base}. ΔT_{s-w} is a log mean temperature difference between the external surface and fluid, defined as [11]:

$$\Delta T_{s-w} = \frac{(T_s - T_{w,in}) - (T_s - T_{w,out})}{\ln \dfrac{T_s - T_{w,in}}{T_s - T_{w,out}}} = \frac{T_{w,out} - T_{w,in}}{\ln \dfrac{T_s - T_{w,in}}{T_s - T_{w,out}}} \tag{5}$$

The thermal resistance, R is determined as:

$$R = \frac{\Delta T_{s-in}}{q} \tag{7}$$

During testing, it was observed that measurements of the outlet fluid temperature demonstrated significant temporal variation, attributed to incomplete mixing of the exiting fluid. It was therefore determined that the mixed outlet temperature could best be estimated indirectly by the use of the energy balance:

$$q = \dot{m} c_p (\mathrm{T_{w,\,out}} - \mathrm{T_{w,\,in}}) + \mathrm{q_{loss}} \tag{6}$$

A zero flow case was tested and losses were seen to be <2 Watts in all cases. Extraneous losses were therefore determined to be negligible when compared to the order of the heat throughput during tested flows.

Other useful non-dimensional numbers were defined as follows:

Reynolds Number:

$$\mathrm{Re}_d = \frac{rud}{m} \tag{8}$$

Nusselt Numbers:

$$\overline{Nu}_{wet} = \frac{U_{wet} d}{k_w} \tag{9}$$

$$\overline{Nu}_{base} = \frac{U_{base} d}{k_w} \tag{10}$$

Prandtl Number:

$$\mathrm{Pr} = \frac{m\, c_p}{k_w} \tag{11}$$

Graetz Number:

$$Gz = \frac{\mathrm{Re}_d \, \mathrm{Pr}\, d}{L} \tag{12}$$

The Nusselt number in a single channel was compared to the classical solution for the hydrodynamically developed but thermally developing case by Hausen, as given by Incropera and DeWitt [11]:

$$\overline{Nu}_{wet} = 3.66 + \frac{0.0668(d/L)\,\mathrm{Re}_d \,\mathrm{Pr}}{1 + 0.04[(d/L)\mathrm{Re}_d \,\mathrm{Pr}]^{2/3}} \tag{13}$$

and to the combined entry length solution due to Sieder and Tate, as presented in [11]:

$$\overline{Nu}_{wet} = 1.86\left(\frac{\mathrm{Re}_d \,\mathrm{Pr}}{L/d}\right)^{1/3}\left(\frac{m}{m_s}\right)^{.14} \tag{14}$$

Hydrodynamic Characterization

An important consideration in the development of cost effective heat exchangers is the pressure required to force a given flow rate. It is therefore desirable to be able to evaluate the pressure head loss based on physical characteristics of the heat sink alone. To this end, the SiC samples were tested and compared to expected pressure drop values as a function of mass flow rate. Pressure drop is best compared by expression in terms of the non-dimensional friction factor:

$$f = \frac{2\Delta P}{r u_m^2}\cdot\frac{d}{L} \tag{15}$$

where u_m is the average channel velocity. Since flow is laminar for all samples at all flow rates in this study, the present results can be compared to the classical solution for Moody friction factor in the developing flow regime for a circular duct as presented for example by Shah and London [12]:

$$f = \frac{\dfrac{3.44}{(x^+)^{1/2}} + \dfrac{1.25/(4x^+) + 16 - 3.44/(x^+)^{1/2}}{1 + 0.00021(x^+)^{-2}}}{0.25\,\mathrm{Re}_d} \tag{16}$$

where x^+ is a non-dimensional length:

$$x^+ = \frac{L}{d\,\mathrm{Re}_d} \tag{17}$$

This correlation converges to the classical solution of $f = 64/\mathrm{Re}_d$ for the case where x^+ is large and the influence of the hydrodynamic entrance length diminishes.

Measurement Uncertainty

All temperature measurements were gathered utilizing type K thermocouple manufactured by Omega engineering. Error for similar wire was previously determined to be ±0.1 C. The primary parameters contributing to the uncertainty in the hydrodynamic analysis were found to be uncertainty in volumetric flow rate, pressure drop, and average hole diameter. The supplier's estimated error in volumetric flow measurement was 2% of full scale, accounting for no more than 6% error in application. Absolute error in pressure measurements were assumed to be 3 mm H_20, introducing large error estimates at lower mass flow rates, with values approaching 40%. Uncertainty in the determination of the average diameter proved to be the largest contributor at higher Reynolds numbers accounting for up to 4% deviation. Propagation of error using the root-sum square method [13], results in a maximum error for thermodynamic measurements of 5% and error in friction factor ranging from 6-40%. The uncertainty intervals shown in Figure 11 are representative of the uncertainty in all of the hydrodynamic data.

RESULTS AND DISCUSSION

Figure 4 shows the overall sample thermal resistance as a function of total mass flow rate collected on a high performing, single row SiC sample, compared to the copper validation sample. For reference, a line representing typical results from a high performing air impingement heat sink is also presented [14]. At modest flow rates, both milli-channel sinks perform significantly better than their air counterparts with the copper and SiC heat sinks yielding thermal resistances of 0.18 C/W and 0.22 C/W respectively, at a flow rate of 507 ml/min. It is interesting to note that despite its lower thermal conductivity, the SiC heat sink compares favorably with the copper heat sink.

The thermal data from these single row samples are validated in Fig. 5 by comparison with classical solutions for single tubes with a constant wall temperature. In this experiment, the surfaces were not held at a fixed temperature. A variation of 5 C at maximum was accounted for by averaging the thermocouple measurements taken along the sample surface. In the plot, data are seen to lie between the Nusselt number solutions for a fully developed hydrodynamic and thermally developing channel flow, and a combined entry length solution, both of which converge to the developed value of 3.66. Since the copper sample has larger holes than the SiC sample, it is expected that the copper sample will tend more to the combined solution, and the SiC to the thermally developing solution.

As additional rows are added to the heat exchanger, the presentation developed for Fig. 5 can be utilized to show that additional physics are in play. Figure 6 shows diminishing values for Nu_{wet} as the number of rows increase. Nu_{wet} compares well to theory for the case of a single row, implying that the wall temperature is nearly constant around the wetted-area and equal to the temperature measured at the outer surface area. When the number of rows increases, the temperature of the internal channels falls below the values measured on the outer surface. This results in an apparent reduction in Nu_{wet}. The diminished values of Nu_{wet} in Fig. 6 as the number of rows increase, affirms that the outer surface temperature and heat flux are not sufficient to describe the interior conditions. Furthermore, the reduction in Nu_{wet} does not imply that the overall performance is diminished in multiple row heat sinks, only that other formulations are necessary to describe their performance.

A common method for evaluation of heat sinks is the overall thermal resistance. A comparison of thermal resistance for samples of four and seven rows is shown in Fig. 7. These samples have identical geometrical parameters except for the number of rows. Here we see that the resistance drops for increase in mass flow rate and for an increase in number of rows, reflecting a corresponding reduction in device temperature. This effect can be captured in a non-dimensional form by introducing an alternative definition of the Nusselt number, Nu_{base}, as formulated in Eq. 10. In this form, the base area is used to describe the overall heat transfer. Figure 8 compares Nu_{base} for samples with four rows and seven rows, and shows an increase in the Nu_{base} with an increase in the number of rows. Here it is seen that multiple rows allow heat to be conducted to subsequent interior rows. Due to the finite conductivity of the SiC samples, Nu_{base} will approach a limit where a further increase in the number of rows will no longer effect the overall thermal performance of the heat sink.

As important as thermodynamic performance in the selection of a viable compact heat exchanger, are the hydrodynamic characteristics. As seen in Fig. 9, the pressure drop from samples of similar cross sectional area depends on both channel diameter and mass flow rate. Data reveal a well-behaved linear relationship between mass flow rate and pressure drop, an expected behavior in laminar flow conditions. Also, a reduction of channel diameter size from 1.43 mm to 0.34 mm, results in a nearly tenfold increase in pressure drop across the sample. Calculated friction factor data for the copper validation sample are shown in Fig. 10, along with hydrodynamically developing friction factor theory. The copper sample contains a set of identically sized, smooth channels allowing for validation of experimental set up for pressure drop measurement. The channels in the copper sample have a relatively small L/d ratio, making entrance and exit losses numerically significant. In Fig. 10, data for f are shown after subtracting inlet and exit pressure losses computed using a rounded entrance loss coefficient, and a contraction/expansion ratio of 0.1. As seen in the figure, these data compare favorably with the theory.

The SiC samples exhibit departures from those idealized in the copper sample. Due to processing and sintering under high temperature and pressure, the SiC samples tested showed varying degrees of variation in channel size and shape. A view of the cross section of a typical sample and its measured friction factor data are shown in Fig. 11. Using the average channel diameter, it is evident that even with irregularity, the friction factors line up well with theory when using hydraulic diameter as the representative length scale. Figure 12 shows friction factor data for samples 3 through 7, again demonstrating good agreement with the theory.

Evidence of Porous Media Behavior

The inability of single channel models to predict the behavior of multi-row heat sinks, demonstrated in Fig. 14, emphasizes the need for models that couple adjoining rows. These models are under development and will be reported in a future publication. To preliminarily examine their behavior, we observe that if an equilibrium porous media formulation is adopted for the entire heat sink, then the interior of the heat sink appears to flow with a uniform slug flow. Heat is diffused in the direction normal to the flow through an effective thermal conductivity that weights the conductivity of both the continuous SiC phase and the discontinuous water phase.

As a first estimate, this weighted conductivity was derived from a "windowpane" model as shown in Fig. 13. The figure shows one quarter of a representative unit cell where resistance to heat flow of the fluid and adjacent solid are in parallel, and both are in series with the lower portion of the unit cell. For the case in which channel and cell widths are equal in x and y, the solution can be represented in terms of porosity, rather than explicit length scales:

$$k_{e,y} = \frac{k_s\left[k_w + k_s\left(\frac{1}{\Phi^{1/2}} - 1\right)\right]}{k_s + \left(1-\Phi^{1/2}\right)\left[k_w + k_s\left(\frac{1}{\Phi^{1/2}} - 1\right)\right]} \tag{18}$$

where $\Phi^{1/2} = a/b$.

Without derivation, it can easily be shown that the temperature solution in the "porous" heat sink will yield a Nusselt number solution of the form:

$$\overline{Nu}_{base} = f\left(\frac{L/H}{\mathrm{Re}_H \ \mathrm{Pr}}\right) \tag{19}$$

where:

$$\overline{Nu}_{base} = \frac{U_{base}H}{k_{e,y}} \tag{20}$$

and:

$$\mathrm{Re}_H = \frac{r\hat{u}H}{m} \tag{21}$$

where:

$$\hat{u} = \frac{\dot{m}}{rHw} \tag{22}$$

Note that the thickness of the entire sample, H, is a more suitable length scale for description of the overall sample geometry. Figure 14 is an attempt to cross-compare all samples using a porous media point of view. It is important to note that the effective conductivity, $k_{e,y}$, were not measured, rather they were calculated from Eq. 18, wherein the porosity was estimated by inspection of photographs of the sample's cross section.

There may be significant error due to the simplicity of the "windowpane" model.

The results of Figure 14 do not support the view that these heat sinks can be described using a porous media approach. This conclusion is stated with caution, since data from some of the heat sinks are anomalous, and effective conductivities need to be measured to confirm the validity of the simple conductivity model. Work in this area is in progress.

CONCLUSIONS

Although a relatively small set of samples were characterized, some definite conclusions can be stated:

1. Liquid-cooled SiC heat sinks easily outperform air-cooled heat sinks and compare favorably with copper equivalents.
2. Performance of single row SiC samples is not hindered by low thermal conductivity and can be predicted by theory developed for isothermal channels.
3. Thermal performance for multiple row samples cannot be captured by single channel, isothermal theory. However, through R and Nu_{base}, multiple row samples are shown to be more effective than single row counterparts.
4. Hydrodynamic performance in the samples agrees with theory and does not suffer from physical irregularities (size and shape variation).
5. Simple porous media model approach so far yields inconclusive results.

Sample ID	L x w x H m $\cdot 10^{-3}$	d m $\cdot 10^{-6}$	No. of rows	N
1	33x22x11.2	1300	5	42
2	33x22x10.8	335	11	155
3	32x22x11.8	940	7	49
4	30x22x5	1150	1	10
5	33x22x5	510	7	126
6	33x22x2.8	510	4	76
7	32x25.4x3.8	2030	1	6

Table 1. Geometric characteristics of tested samples.

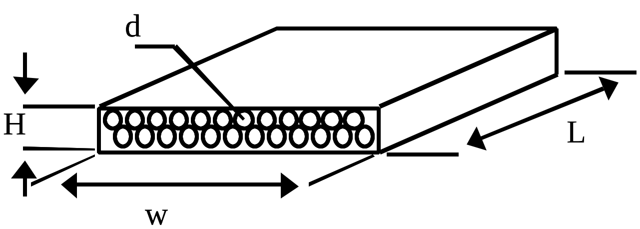


Figure 1. Schematic of a representative sample

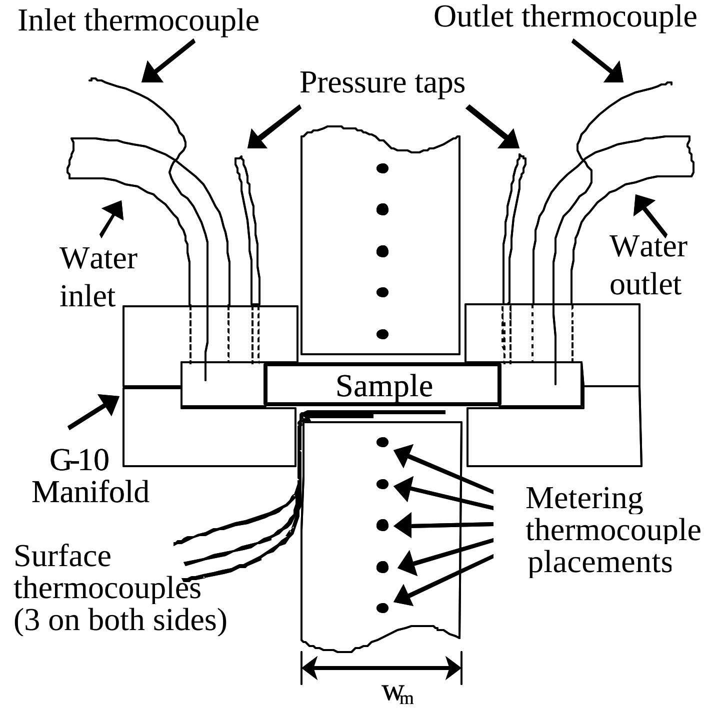
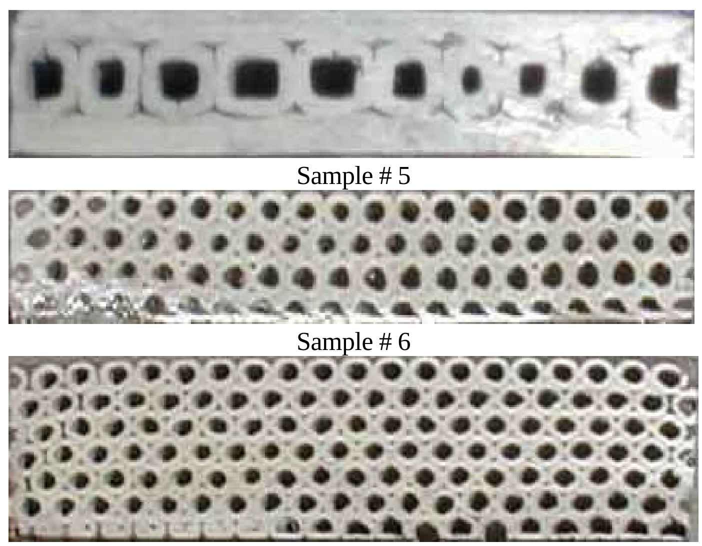


Figure 2. Schematic of sample manifold and instrumentation



Sample # 5

Sample # 6

Sample # 7

Figure 3. Photographs of representative samples

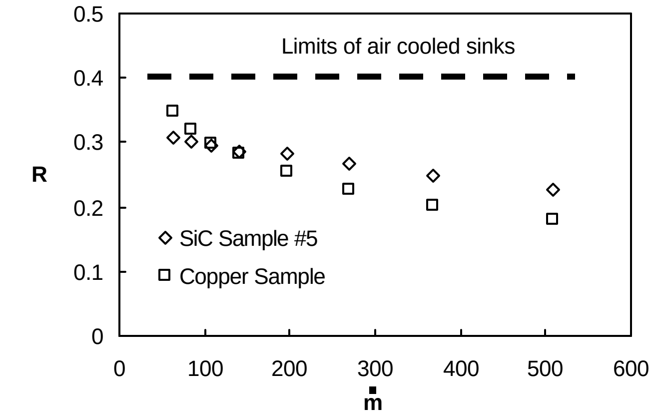


Figure 4. Comparison of thermal resistance for tested SiC and Copper heat exchangers and high performance air sink.

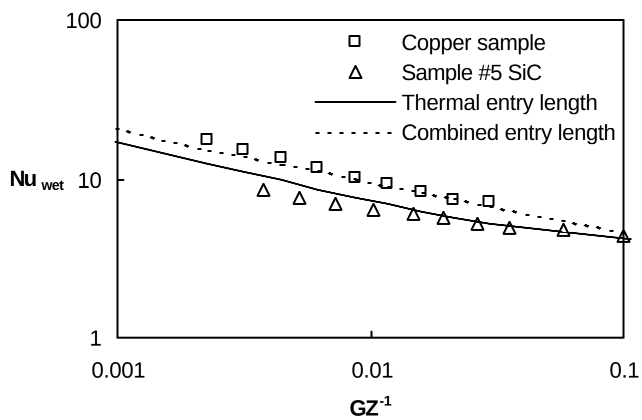


Figure 5. Single row samples compared to classical heat transfer theory.

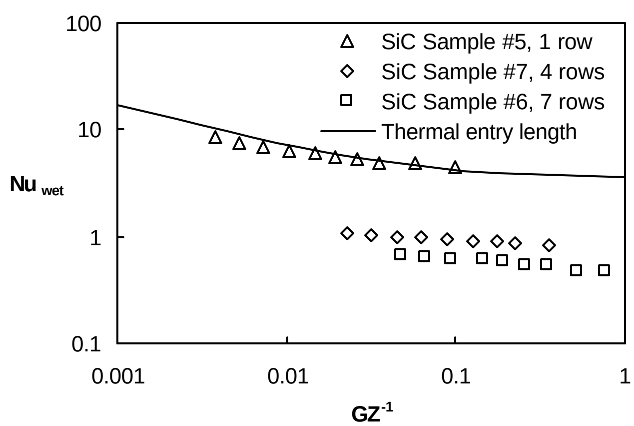


Figure 6. Wetted Nusselt number as varies with number of rows

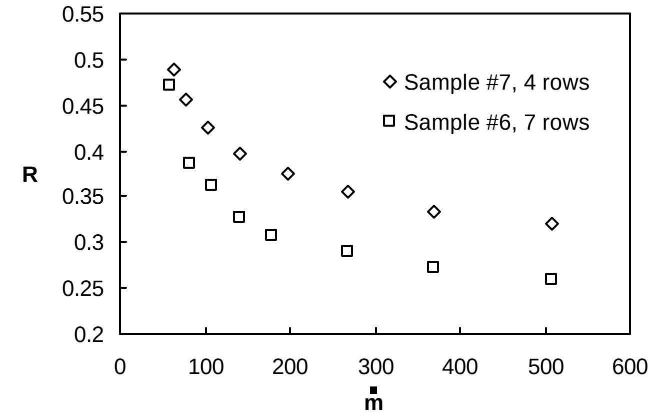


Figure 7. Effect of increasing number of rows on thermal resistance.

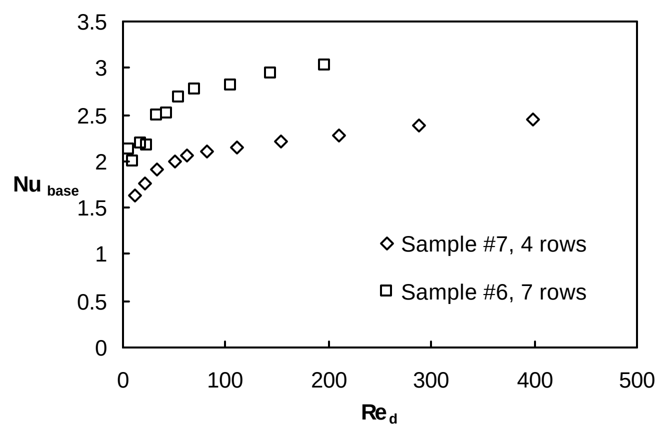


Figure 8. Non-dimensional data comparison.

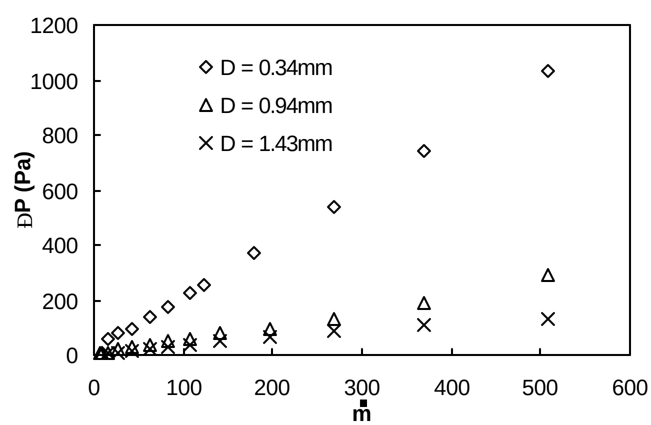


Figure 9. Pressure drop as a function of mass flow and channel diameter for samples of similar cross sectional area.

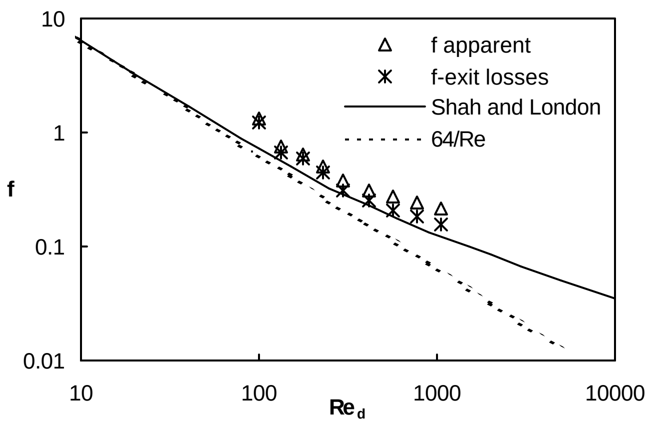


Figure 10. Friction factor for the copper validation sample as compared to theory.

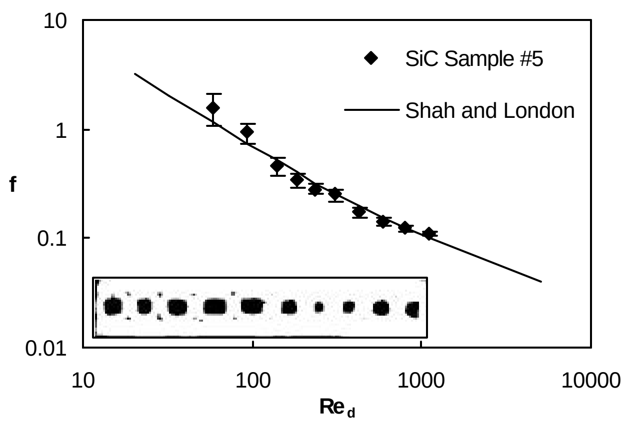


Figure 11. Friction factor for a representative SiC sample as compared to theory.

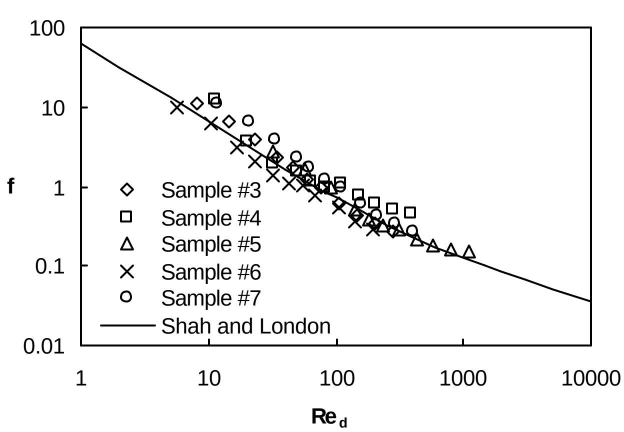


Figure 12. Friction factors of several SiC samples compared to theory.

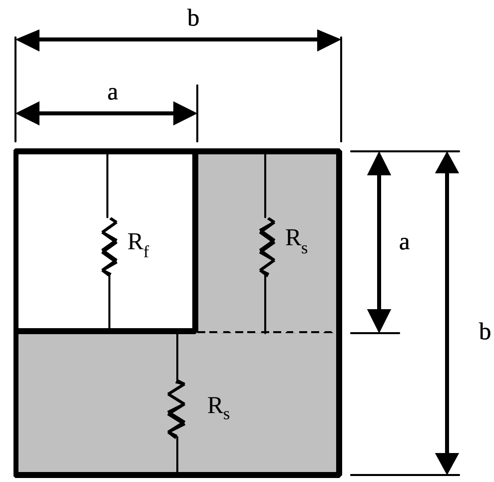


Figure 13. Schematic of quarter symmetry window pane model used in determination of effective y-direction conductivity, $k_{e,y}$

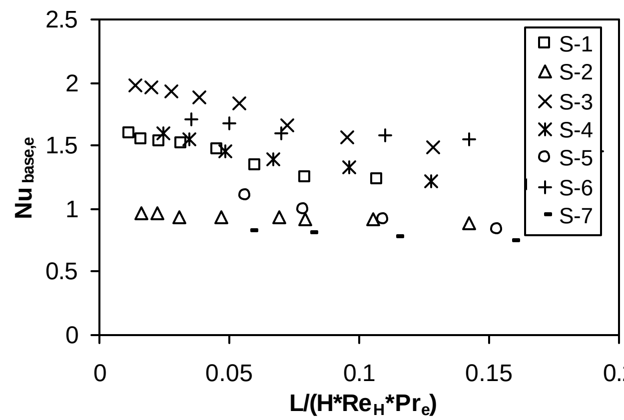


Figure 14. Nusselt number based on porous media approach for all samples.

ACKNOWLEDGMENTS

The research was performed under Army STTR contract no. DASG60-02-P-02-P-0280 monitored by Richard Rodgers.

REFERENCES

[1] Ortega, A., 2003, "Air cooling of electronic components: review of research developments in the period 1981-2001 and their impact on our understanding of cooling limits", 2003 ASME IMECE, Washington D.C. Nov. 2003, paper # IMECE 2003-42412

[2] Tuckerman, D.B., and R.F.W. Pease, 1981, "High-performance heat sinking for VLSI", IEEE Electron Dev. Lett. 2, pp. 126-129

[3] Phillips, R.J., 1990, “Micro-channel heat sinks, in A. Bar-Cohen A.D. Kraus (Eds.)”Advances in Thermal Modelling of Electronic Components and Systems vol. 2, ASME, New York, Chapter 3

[4] Lu, T.J., and M.F. Ashby, 1998, “Heat transfer in open celled metal foams”, Acta Mater, **46**, pp. 3619-3635

[5] Knight, R.W., and J.S. Gooding, D.J. Hall, 1991, “Optimal thermal design of forced convection heat sinks-Analytical” ASME Journal of Electronics Packaging, **113**, pp. 313-321

[6] Koh, J.C.Y., and R. Colony, 1986, “Heat transfer of microstructure for integrated circuits”, Int. Comm Heat Mass Transfer, **13**, pp. 89-98

[7] Kim, S.J., D Kim, and D.Y. Lee, 2000, “On the local thermal equilibrium in microchannel heat sinks” Int Journal of Heat Mass Transfer, **43**, pp. 1735-1748

[8] Kim, S., J. D Kim, 1999, “Forced convection in microstructure for integrated circuits” ASME J. Heat Transfer, **121**, pp. 635-645

[9] Mudawar, I., 2000, “Assessment of high heat flux thermal management schemes”, IEEE Inter Society Conference on Thermal Phenomena

[10] Vafai, K., and L. Zhu, 1999, “Analysis of two-layered micro-channel heat sink concept in electronic cooling”, International Journal of Heat and Mass Transfer, **42**, pp. 2287-2297

[11] Incropera, F., and D. Dewitt, 2002, “Fundamentals of heat and mass transfer” 5^{th} ed., New York, Wiley, pp. 482-490

[12] Shah, R.K., and A.L. London, 1978, Laminar flow forced convection in ducts, (Supp. 1) in advances in heat transfer, in: T.F. Irvine, J.P. Hartnett, Academic, New York, pp. 96-98

[13] Moffat, R.J., “Describing the uncertainties in experimental results, Exp. Thermal Fluid Sci. 1 (1988) 3-17.

[14] Issa, J.S., Ortega, A., “Experimental Measurements of the Flow and Heat Transfer of a Square Jet Impinging on an Array of Square Pin Fins,” Proc., IMECE2002, ASME International Engineering Congress & Exposition, New Orleans, Louisiana, Paper # 34244.

